

INDIAN STATISTICAL INSTITUTE

Midsemester Examination

M Math I Year

Algebra I

Date: September 11, 2024 Maximum marks:30 Duration: 90 minutes

1. Let R be a commutative ring with 1 and M be an R -module. Consider the set

$$\text{Tor}(M) := \{m \in M \mid rm = 0 \text{ for some nonzero } r \in R\}.$$

- (a) If R is an integral domain, then show that $\text{Tor}(M)$ is a submodule of M . [3]
(b) If R is not an integral domain, show that $\text{Tor}(M)$ may not be a submodule of M . [3]

2. Let R be a commutative ring with 1 and M be an R -module. Let A, B, C be submodules of M such that $C \subseteq A$. Prove that [6]

$$A \cap (B + C) = (A \cap B) + C.$$

3. Check the irreducibility of the following polynomials:

- (a) $X^4 + 1$ in $\mathbb{Z}[X]$. [3]
(b) $(X - 1)(X - 2) \cdots (X - n) - 1, n \geq 1$, in $\mathbb{Z}[X]$. [4]

4. Let R be a UFD. Show that $R[X]$ is also a UFD. [8]

5. Let $p \in \mathbb{Z}$ be a prime. Show that the ideal $\langle p, X \rangle$ of $\mathbb{Z}[X]$ is not principal. [5]